

Freight Rate Prediction

- A CatBoost model predicting posted_rate/distance, combined with a parametric date effect that extrapolates the market trend into the ungraded forecast window.
- Evaluated on a withheld forward holdout that Optuna never saw.

Headline result (holdout 2025-09-01 → 2025-10-31, 9,523 loads)	Value
MAPE — all rows (what score.py grades)	4.653 %
MAPE — excluding corrupted labels	2.163 %
MAE	\$107.17
RMSE	\$633.95
Median APE	1.835 %
Bias	−0.30 %
R ²	0.8274
Rolling-origin CV, 4 folds	4.430 % (sd 0.229)

1. Key findings from exploring the data

1.1 Rate is almost perfectly explained by distance

On the raw data correlation between distance and posted_rate is 0.9085. After cleaning the corrupted labels (Section 2.1), it became 0.9889, and the log-log correlation is 0.9935. Other numeric columns, weight, market index and quote signal all sit below ± 0.05 .

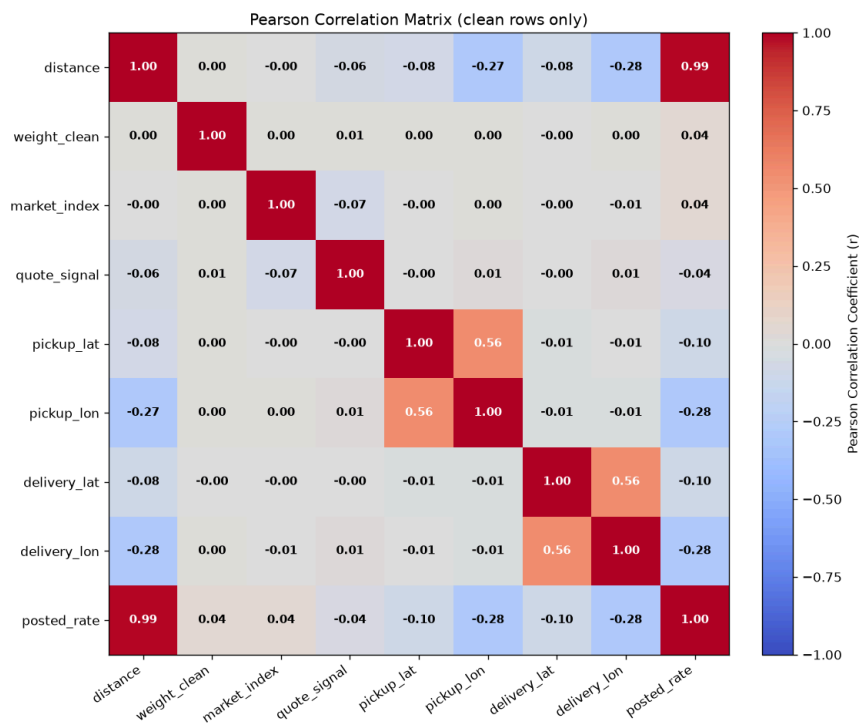


Figure 1 : Pearson correlations on clean rows. Distance dominates; every other numeric column is near zero.

1.2 Market index is a national daily figure

Date explains 97.8 % of the variance in market_index, within-day standard deviation is 0.025 against 0.168 overall. This makes it recoverable when missing and reconstructable for December, whose input file has no such column.

1.3 Eight cities appear only in validation

Using categorical data depending on place is useless here. Their coordinates are known, so lanes are represented geometrically instead.

2. Data-quality issues and how they were addressed

2.1 677 corrupted posted_rate labels (1.41 %)

posted_rate has a maximum of 25,533 against a median of 2,031, and a skew of 1.90. The target deserves the same scepticism as the features

Comparing each load's posted_rate/distance against the median for its peer group — distance ventile $\times$ equipment, produces a distribution that is not a heavy tail but three separated populations.

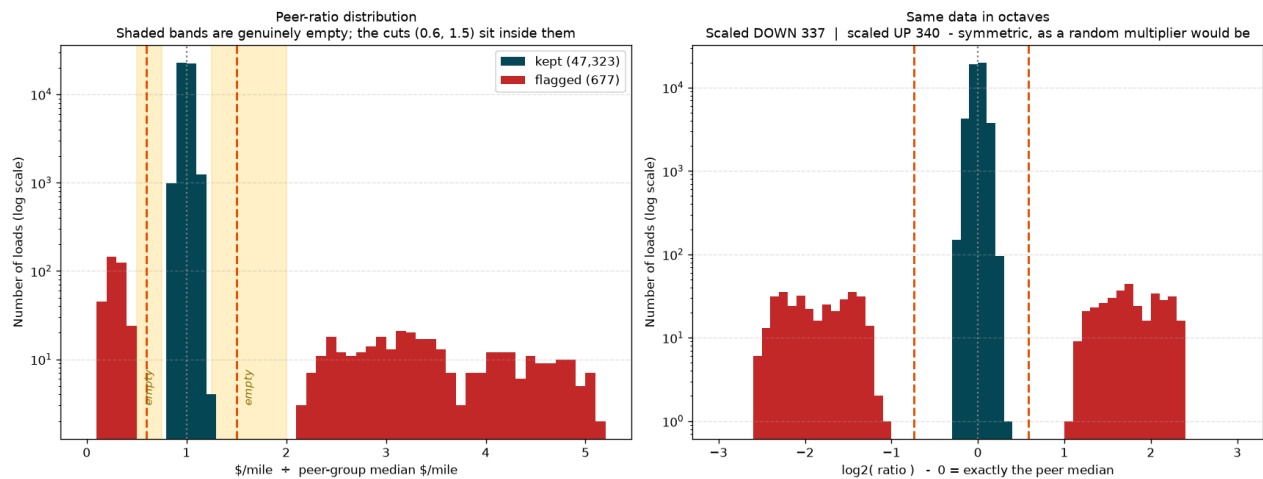


Figure 2 : Peer-ratio distribution (log y-axis). Shaded bands are genuinely empty; the cut points sit inside them. Right: the same data in octaves — 337 labels scaled down, 340 scaled up.

Three properties identify this as injected corruption rather than genuine price variation:

- The histogram has genuinely empty gaps at [0.50, 0.75) and [1.25, 2.00). Every threshold inside those gaps selects the identical 677 rows, so the cut is not tuned.
- The flagged rows split 337 scaled down against 340 scaled up - symmetric in log space, the signature of a random multiplier. A genuine heavy tail is one-sided.
- They are spread evenly across all ten months (~67 per month), ruling out a period-specific data problem.

Solution : these rows are dropped from model fitting only, never from anything scored. The grader evaluates all 12,000 validation loads.

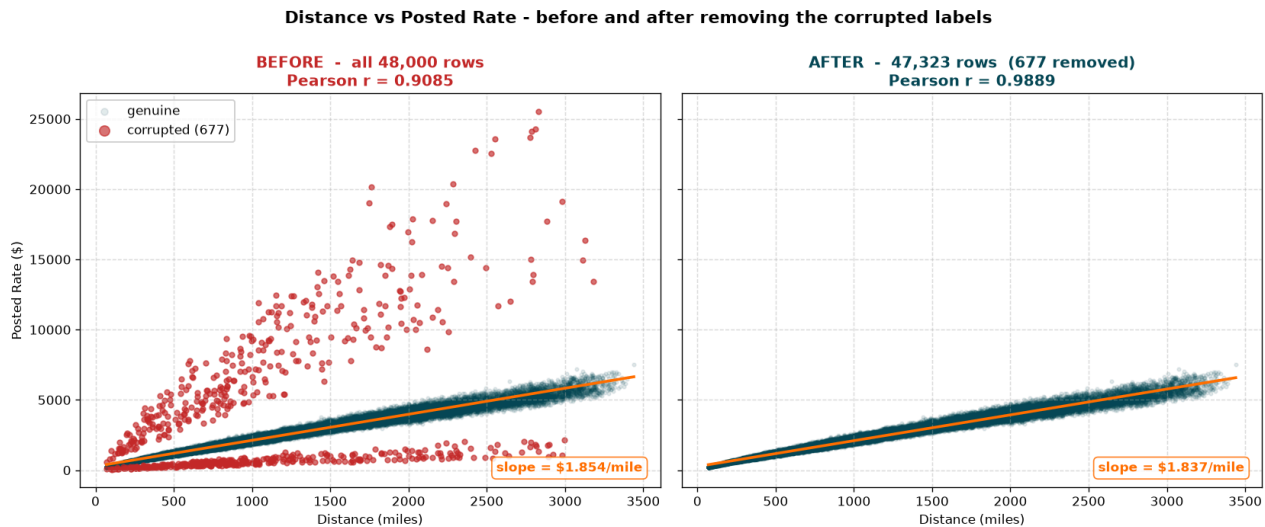


Figure 3 : Distance against posted rate, before and after excluding the corrupted labels. Pearson r rises from 0.9085 to 0.9889; skewness falls from 1.90 to 0.69.

2.2 292 negative weights

Weight ranges down to $-47,500$. The magnitudes are entirely plausible and only the sign is wrong, so these are data-entry defects rather than an outlier population. Applying `abs()` recovers all 292 rows instead of discarding them.

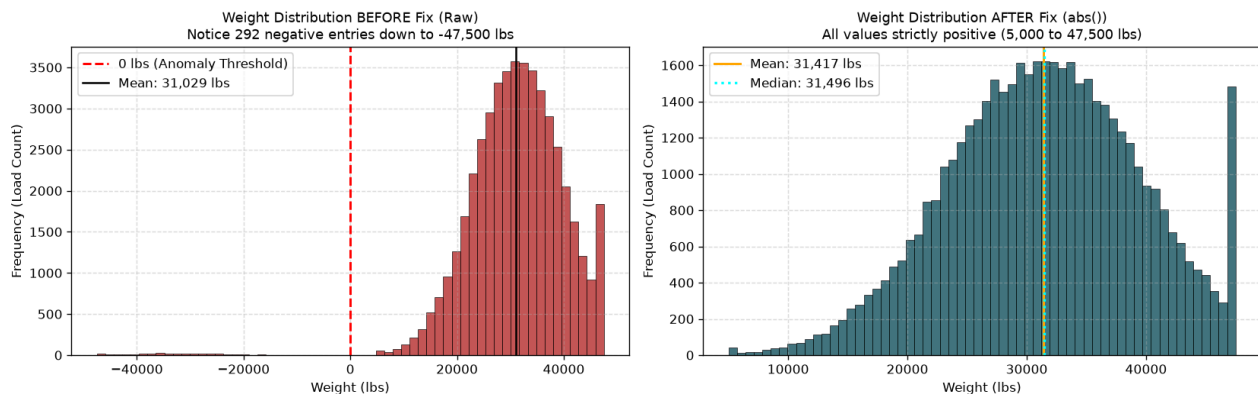


Figure 4: Weight distribution before and after sign repair. The corrected range, 5,000–47,500 lbs, is entirely plausible.

2.3 Missing values: 300 weights, 374 market indices

Since the missing percentage is very low (weight: 0.62%, market_index: 0.78%) dropping those rows is also feasible. But both were imputed rather than dropped.

- Weight is filled from the median for its equipment type
- market_index from the daily calendar, which recovers it to within ± 0.025 given that date explains 97.8 % of its variance.

Each carries an indicator column (weight_imputed, market_index_imputed) so the model can condition on missingness. Imputation statistics are fitted on development data only and then applied unchanged to validation and December, so no information flows backwards.

2.5 quote_signal is unusable

The quote_signal column is absent from december-chart-inputs.csv entirely, and score.py rejects the file if its columns change. Using it would mean inventing a value for a third of the required predictions. Its correlation with posted rate is -0.04 , so nothing is lost by dropping it.

3. Reasoning behind the chosen model

3.1 Why a gradient-boosted model

The problem is tabular regression on 48,000 rows with 25 mixed numeric and categorical features, strong non-linearity (price per mile falls with distance, and geography interacts with equipment) and no sequence or image structure. Gradient-boosted trees are the default for that shape of problem, and the alternatives were measured rather than dismissed. All four rows below predict the same posted_rate/distance target on the same 17 features and are scored on the same withheld holdout.

Model family	Holdout MAPE	Why it falls short
Ridge regression (linear)	7.044 %	Cannot represent the geography $\times$ equipment $\times$ distance interactions
Neural network (2 $\times$ 64 MLP)	6.476 %	48,000 rows is small for a network; needs scaling and far more tuning
Random forest (300 trees)	5.161 %	Competitive, but averages trees rather than fitting residuals
Gradient boosting (shipped)	4.653 %	—

3 properties made gradient boosting the right family here, beyond the headline number:

- It is invariant to monotone transformations of a feature, so no scaling, normalisation or skew correction is needed.
- It handles mixed numeric and categorical inputs natively, which matters because equipment is the one genuinely useful categorical and the geography is continuous.
- It fits residuals sequentially, which is exactly the structure the hybrid in Section 3.3 needs: the booster is repeatedly refitted against a target with the date effect removed. A random forest averages independent trees and does not compose that way as naturally.

The one place a boosted tree is the wrong tool is extrapolation — it cannot predict beyond the range of dates it was trained on. That limitation is handled by pairing it with an ordinary least-squares term, described in Section 3.3. The model uses each method where it is strong: trees for the cross-section, a straight line for time.

3.2 Target: dollars per mile, not dollars

The model predicts `posted_rate / distance` and multiplies the known distance back in. Distance is a given input in all three files, so this reconstruction is exact and introduces no bias.

The reason is specific, and it is not that the tree saves effort. As a plain booster, raw dollars and dollars per mile are indistinguishable (5.291 % against 5.288 %). The ratio matters because of the date effect described next, which assigns one number per day to every load on that day. A +1.28 % market drift is worth \$7.60 on a 227-mile load and \$63.26 on a 2,606-mile load - an 8.3× spread that no single constant can represent, injecting about 0.94 pp of error. Expressed in dollars per mile the same drift spans only 1.4×, which one number fits. Inside the hybrid the ratio is worth 0.71 pp; on raw dollars the date effect actually performs worse than no date effect at all.

3.3 Structure: boosted trees plus an extrapolating date effect

The graded window is November and December; training ends 31 October. A gradient-boosted tree cannot extrapolate - it reproduces the last date bin it saw and then flattens, which shows up as a persistent low bias on any forward holdout.

The model therefore fits two components by backfitting.

- A date-free booster captures geography, distance, weight and equipment.
- The daily residual is then regressed by ordinary least squares on elapsed days, market index and six day-of-week indicators- a straight line extrapolates, so the trend survives into the forecast window.

The booster is refitted against a target with that date effect removed, so the two halves stop competing for the same variation.

Annual seasonal terms were tested and rejected. The window spans 303 days - 0.83 of a year, so an annual cycle cannot be separated from a trend. Adding sine and cosine of day-of-year cost 0.59 pp on the holdout and was worse on three of four folds.

The drift term is only identified when elapsed time and market index are not collinear. They are strongly collinear early in the year (+0.88 for January–May), and the code measures this precondition before trusting the term, falling back to the plain booster above a threshold of 0.55. Over the full development window the value is −0.109, so the drift is well identified for the shipped fit.

3.4 Feature engineering

Twenty-five features are built, of which the shipped hybrid uses seventeen (the calendar and market columns are owned by the date effect instead). One constraint governs all of them: every feature must be reconstructable from the six columns `december-chart-inputs.csv` supplies ; pickup, delivery, distance, equipment, weight and date. A feature that cannot be rebuilt for December is not a feature, however predictive.

Technique	Features produced	Rationale
Lookup reconstruction	<code>pickup_lat/lon</code> , <code>delivery_lat/lon</code> , <code>market_index</code>	December ships no coordinates and no market column; a city→coordinate table and a date→index calendar rebuild both
Great-circle geometry	<code>haversine_mi</code>	True straight-line distance between endpoints, independent of reported mileage

Technique	Features produced	Rationale
Ratio feature	$\text{circuitry} = \text{distance} \div \text{haversine_mi}$	Normally ~ 1.18 ; spikes to 1.71 on loads clipped to the 70-mile floor, so it flags distorted mileage
Corridor midpoint	mid_lat, mid_lon	Stands in for the lane corridor without needing a lane identifier
Directional deltas	delta_lat, delta_lon	Separates a westbound haul from its eastbound return, which price differently despite sharing a midpoint and distance
Cyclical encoding	bearing_sin/cos, dow_sin/cos, doy_sin/cos	Direction and calendar position are circular; sine–cosine pairs stop 359° and 1° (or Sunday and Monday) being treated as far apart
Threshold flag	is_short_haul	Marks loads at the observed 70-mile distance floor as their own regime
Missingness indicators	weight_imputed, market_index_imputed	Lets the model condition on the fact a value was filled rather than observed
Ordered categorical code	equipment_code	Dry Van / Flatbed / Reefer as a 3-level code, handled natively by each backend
Target ratio	$\text{posted_rate} \div \text{distance}$	Not an input feature but the modelled quantity

The central design decision is to represent a lane by its endpoint geometry rather than by its name. A categorical lane or city feature has no slot for the eight cities that appear only in validation, 12.1 % of the graded set, whereas coordinates place any city on the map whether or not it was seen in training. Section 1.5 shows the identity encoding also carries no measurable signal once its cardinality is accounted for, so nothing is given up by omitting it.

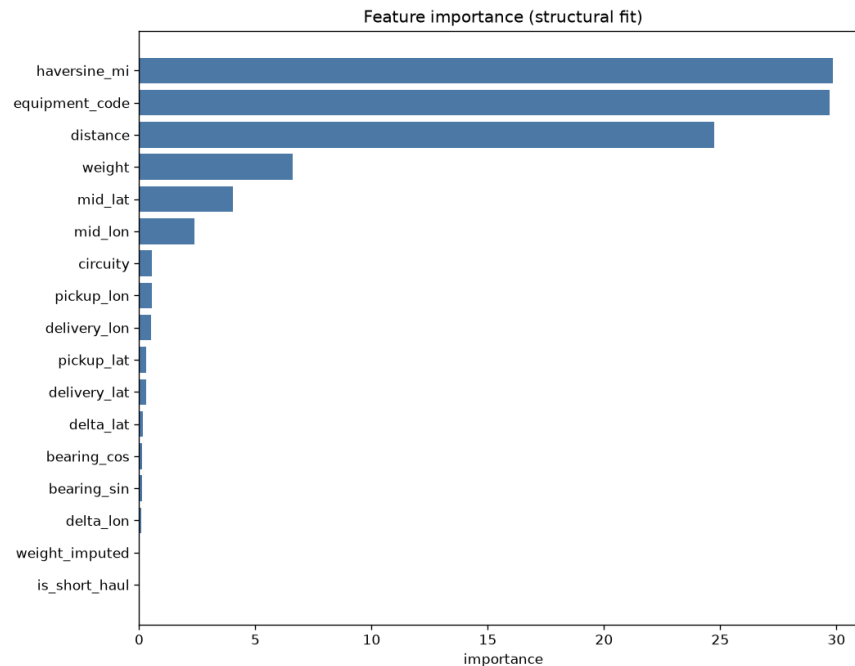


Figure 5: Feature importance of the shipped model. Geography and distance dominate; equipment is the one categorical that earns its place.

3.5 Backend: chosen by measurement

All four gradient-boosting libraries were compared at matched capacity - identical learning rate, round count, leaf count, minimum leaf size and L2 penalty ,across two model shapes and two encodings, each scored by the same rolling-origin cross-validation. Sixteen configurations in total.

Configuration	CV MAPE	sd	MAE	Seconds
catboost · hybrid · coords	4.495 %	0.270	\$107.52	83
catboost · hybrid · coords+city	4.526 %	0.269	\$108.45	107
xgboost · hybrid · coords+city	4.540 %	0.281	\$108.44	22
lightgbm · hybrid · coords	4.588 %	0.326	\$109.77	6
catboost · boosted · coords	4.638 %	0.468	\$111.16	59
lightgbm · boosted · coords	4.754 %	0.489	\$114.07	5

Representative rows from the 16-cell grid.

Two conclusions follow. Model structure matters roughly twice as much as library choice: every hybrid configuration beats every plain booster by about 0.16 pp, while the spread across all four libraries within the hybrid is 0.077 pp. And CatBoost wins, though narrowly , its margin over XGBoost is within one fold standard deviation, and it costs four to fourteen times more to fit. It was selected on the strength of also having the lowest fold-to-fold variance, which matters for a single graded submission.

4. Training and validation approach

4.1 Why the split is chronological

The graded set is November and December 2025 while development data ends 31 October. Every estimate in this report therefore holds out a trailing window and asks the model to predict forward. A random K-fold would let the model train on October while predicting June - it would score better and would report a number the graded set cannot reproduce.

4.2 Rolling-origin cross-validation

Four origins, each training on everything before the cutoff and testing on the 61 days after it.

Cutoff	Train period	Train rows	Test period	Test rows
2025-06-01	Jan 1 – May 31	24,023	Jun 1 – Jul 31	9,695
2025-07-01	Jan 1 – Jun 30	28,806	Jul 1 – Aug 30	9,524
2025-08-01	Jan 1 – Jul 31	33,718	Aug 1 – Sep 30	9,429
2025-09-01	Jan 1 – Aug 31	38,477	Sep 1 – Oct 31	9,523

4.3 The holdout is withheld from tuning

Optuna tunes against the June, July and August origins only. The 2025-09-01 holdout is never shown to the search, so the headline 4.653 % is measured on data the hyperparameters were never fitted to. Fifty trials searched for learning rate, round count, leaf count, minimum leaf size, L2 penalty, row subsampling and column subsampling; the best tuning score was 4.356 %.

That final fold is also the most informative one. It has the longest training window and the lowest time–market collinearity (+0.349), making it the only fold where the date effect is identified rather than falling back to the plain booster , and the closest structural analogue to the real task, which trains on January–October and predicts November–December.

4.4 Results

Metric	All rows	Clean rows only
MAPE	4.653 %	2.163 %
MAE	\$107.17	\$51.28
RMSE	\$633.95	\$76.37
Median APE	1.835 %	1.807 %
Bias	−0.30 %	—
R ²	0.8274	0.9969
Rows	9,523	9,379

The difference between the two columns is the corruption floor described in Section 2.1, not a modelling choice. The date effect cuts average bias roughly fourfold, from -1.19% to -0.31% , and removes the consistent undershoot, but it slightly over-corrects, leaving a mild positive drift late in the window.

5. Fixed December prediction chart

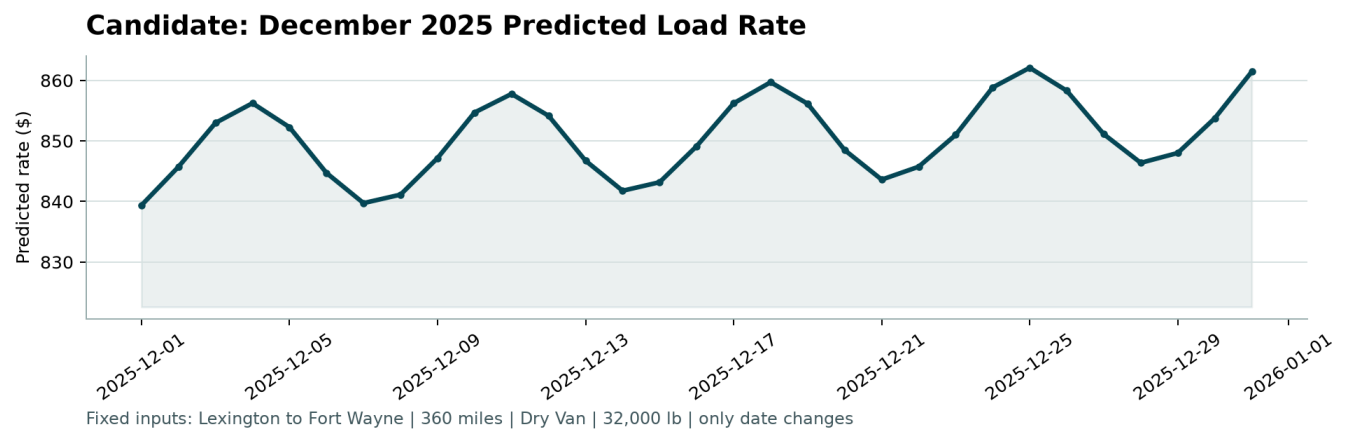


Figure 6 : Predicted rate across December 2025, all inputs fixed except date.

The chart is a direct read-out of the date effect. Two components are visible and both are intended. The seven-day cycle is the day-of-week term: midweek pickups price above weekend ones, learned from six day indicators rather than imposed as a smooth shape. The upward slope from 839 on 1 December to 861 on 31 December is the drift term carrying the market trend forward, combined with the published daily market index.